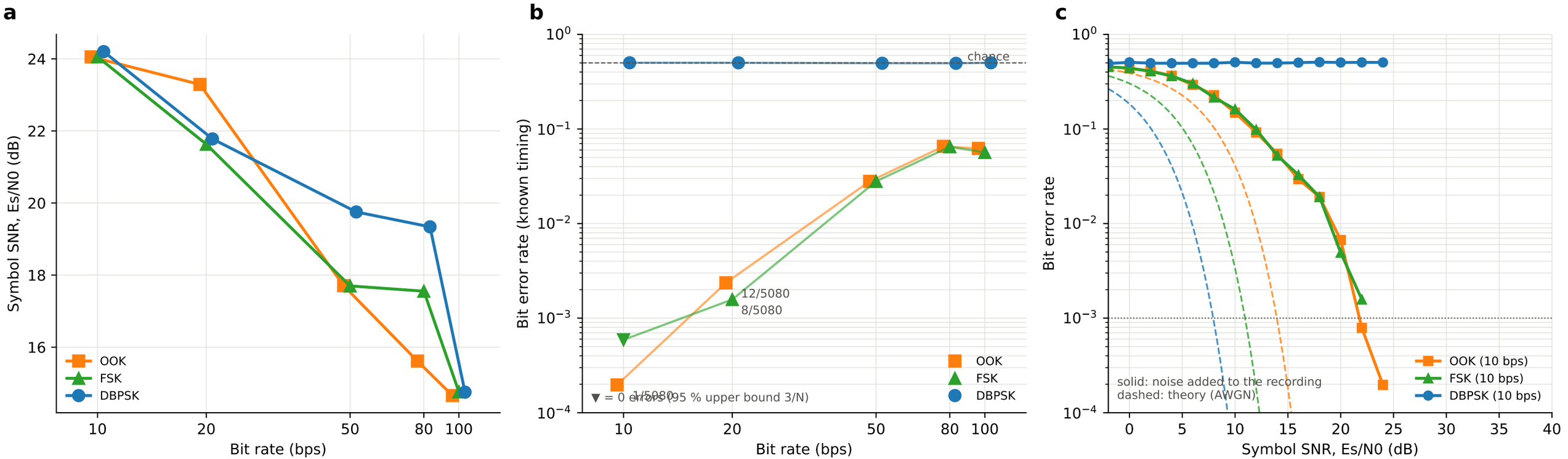
a
Symbol SNR, Es/N0 (dB)
24
22
20
18
16
10
20
50
80
100
Bit rate (bps)
OOK
FSK
DBPSK
b
Bit error rate (known timing)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
chance
12/5080
8/5080
1/5080
▼ = 0 errors (95 % upper bound 3/N)
10
20
50
80
100
Bit rate (bps)
OOK
FSK
DBPSK
c
Bit error rate
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
solid: noise added to the recording
dashed: theory (AWGN)
0
5
10
15
20
25
30
35
40
Symbol SNR, Es/N0 (dB)
OOK (10 bps)
FSK (10 bps)
DBPSK (10 bps)